

ParlLawSpeech

Codebook

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1. Introduction

Parliaments are key institutions of democracy. The documents that parliaments produce - including plenary protocols, legislative bills, and ultimately adopted laws - thus contain invaluable information for understanding how democratic governance works while advances in natural language processing offer an ever increasing potential to extract and analyse this information systematically.

Yet, despite this significance of analysing parliamentary documents systematically, accessing them often involves very notable hurdles in applied projects (Sebők et al., 2025). To conduct computational text analysis effectively and to compare across different legislatures, researchers require comprehensive, consistent, and machine-readable text corpora. *ParlLawSpeech* (PLS) contributes to and extends recent efforts - e.g. the CLARIN project (De Jong et al., 2022), the Comparative Agendas Project (Baumgartner et al., 2019), ParlSpeech (Rauh & Schwalbach, 2020), or Parl-EE (Sylvester, et al., 2022) - to make such corpora available to the research community.



The PLS corpora have been created under [Work Package 5 of the OPTED project](#) (Observatory for Political Texts in European Democracies, Horizon 2020 Grant [951832](#)). Besides extending the coverage of readily available parliamentary text corpora, PLS innovates in two respects. First, it is not constrained to only one type of parliamentary document but rather provides full texts for speeches, bills, and laws. Second, it offers data linkage across these different types of parliamentary texts in a way that is typically not even offered by institutional repositories and data providers (Kiss & Sebők, 2022). These two innovations allow users to map and to combine different parliamentary outputs, thereby enabling them to address research questions spanning the full process of parliamentary decision-making from draft documents, over plenary debates, to the contents of the finally adopted rules. In sum, PLS offers machine-readable full-text vectors of parliamentary speeches, bills, and laws, as well as relevant metadata for seven European countries (Austria, Czechia, Croatia, Denmark, Germany, Hungary, and Spain) and the supranational European Parliament (EP).

The present codebook supports potential users of the data files by providing a description of coverage, data collection, validation, and definitions of variables in the speeches, bills, and law datasets.

2. Coverage and data collection

Table 1 summarises the descriptive statistics of the three corpora available by country. PLS encompasses at least ten years of data for all text types and countries, but the time periods covered vary across legislatures due to availability in the underlying institutional archives of each parliament and the intention to include only completed legislative periods, with the year column denoting either calendar years or the respective government cycles.

The country selection of PLS follows two main rationales: First, we aimed to maximise comparative value by providing data on legislatures across different political systems, age, and regions of the European Union (including the EP itself): Second, we built on already existing precursors (especially ParlSpeech) to minimise adaptation costs while matching the necessary language knowledge available in the project team (cf. Schwalbach and Rauh 2021).

Data collection was implemented with web scraping scripts customised specifically to each individual institutional archive for each text type in each country. The datasets are as

comprehensive as the underlying institutional archives within the time period covered. That is, we collected all full texts and meta data for speeches, bills, and laws that each institutional online archive contains (leading to missing values in cases where data is missing or incorrect on the respective website). The matching of speeches, bills, and laws were done by qualitative analysis, tackling the problem of lack of APIs or common identifiers in the archives.

COUNTRY	DATA	COUNT	VARIABLES	TIME FRAME
Austria	<i>Laws</i>	3030	7	1996-2020
	<i>Bills</i>	5926	11	1996-2020
	<i>Speeches</i>	204881	14	1996-2019
Czech Republic	<i>Laws</i>	844	7	2013-2023
	<i>Bills</i>	2127	6	2013-2023
	<i>Speeches</i>	192979	13	2013-2023
Croatia	<i>Laws</i>	2972	5	2003-2020
	<i>Bills</i>	3676	14	2003-2020
	<i>Speeches</i>	405260	11	2003-2020
Denmark	<i>Laws</i>	3220	7	2007-2022
	<i>Bills</i>	3615	10	2007-2022
	<i>Speeches</i>	716807	9	2007-2022
EP	<i>Laws</i>	10554	6	1994-2024
	<i>Bills</i>	14105	11	1994-2024
	<i>Speeches</i>	574119	17	1999-2024
Germany	<i>Laws</i>	1638	6	2009-2021
	<i>Bills</i>	2445	10	2009-2021
	<i>Speeches</i>	191932	13	2009-2021
Hungary	<i>Laws</i>	4303	6	1994-2022
	<i>Bills</i>	7500	7	1994-2022
	<i>Speeches</i>	452645	13	1994-2022
Spain	<i>Laws</i>	1563	6	1996-2023
	<i>Bills</i>	4188	10	1996-2023
	<i>Speeches</i>	318576	13	1996-2023

Table 1: Descriptive statistics of the three corpora per legislature

3. Data structure and variables

The speeches, bills, and laws datasets are available in .RDS formats for compactness and easier use in the R environment (Corpus_[speeches/bills/laws]_[country].RDS). The variables, if not indicated differently, are UTF-8 encoded. In the following, we provide short descriptions of the variables in the three corpora. It is important to note that most of the differences between the metadata columns stem from the collection processes, as the respective legislatures' conventions were closely followed during the creation of PLS. The “core” and “additional”

variables in the datasets are listed separately in this codebook for easier navigation in the datasets. Core variables are defined as columns that are missing from at most two legislatures.

The most important novelty of the PLS datasets is the unique identifier with which linking through the three corpora is possible. This **procedure_ID**, found in the speeches, bills, and laws datasets alike, is based on the conventions of the respective countries' parliamentary websites or other legislative databases. With the help of this ID, researchers can track which bills became which laws and if there were any speeches given regarding them during the legislative process.

3.1) Speeches

3.1.1) Core variables

The first corpus, presented in Table 2, is composed of the parliamentary speeches in clean text formats (**text**) alongside their respective metadata. The **date** on which the speech was given (in YYYY-MM-DD format) is included in the dataset, ensuring accurate filtration of relevant texts for researchers. The **speaker** variable indicates the name of the person having given the respective plenary speech, while the **party** column holds the speaker's party affiliation (in the case of the European Parliament, the variable refers to the party group and not to the national party affiliation). These could be of use for identifying trends in parliamentary parties' activity, as well as finding topics or themes specific to them. The Czech corpus contains the party affiliation as well as party support of non-MP speakers as well.

The **agenda** items under which the speeches were given are available for all legislatures, holding a string with the title of the respective agenda points. This variable provides context for the speeches, helping identify themes or facilitate comparison between countries. **Speechnumbers** are IDs reflecting the chronological order of the speeches within the given session, providing a way to sort and reference speeches conveniently. This variable is numerical with the exception of the Hungarian dataset: Due to the specific data structure of the Hungarian parliamentary website, in some cases several speeches are assigned to one speaker in a single row. In this case, the respective variable indicates the number of speeches (e.g. "4-6"). Finally, the variable **partyfacts_id** makes it possible to link a speaker's party affiliation to parties in the PartyFacts dataset (Döring & Regel, 2019).

Speeches corpora

.....

speaker	Name of the person having given the plenary speech
text	Full text of the respective speech
party	Party or partisan factor of the speaker
date	The date on which the speech was held
agenda	The agenda item under which the speech was given
speechnumber	Unique ID per session reflecting the chronological order
procedure_ID	The unique PLS identifier linking speeches, bills, and laws
partyfacts_ID	ID that facilitates linkage to core parties in the PartyFacts dataset

Table 2: The core variables of the speeches corpora

3.1.2) Additional variables

Several additional variables can be found in the corpora based on the respective legislature’s official sources. The legislative **period** derived from the dates is included in most cases, either holding the electoral cycle (as in the case of the Hungarian data) or a unique ID of the legislative period (as seen in the Austrian corpus). In some cases, the **session** variable specifies the session number within the legislative period as counted by the legislature (this is especially relevant when more than one session was held per day). The **chair** variable (logical TRUE/FALSE values) indicates whether the speaker was the chair of the given session. It is useful for filtering relevant speeches, as the ones given by the chairs usually only hold the agenda points and the names of the speakers. **Links** to the original speeches are provided in many datasets, as they are essential for traceability and the overall reliability of the corpora.

Raw texts (**text_raw**) were left in the corpora in certain cases to make the text cleaning and preprocessing steps reproducible and traceable, allowing for, among other things,

identifying errors in these procedures. Furthermore, these raw texts and speaker names can contain additional information such as spontaneous interventions during a speech or additional speaker information such as titles. There is also supplemental information on the speakers in some datasets, such as the **chancellor**, **minister**, or **schriftführer** variables, each indicating different positions in the parliaments. In the case of the Czech data, there is a general **position** variable that indicates all labelled positions and a variable that indicates whether a speaker is **MP**.

The Spanish **grupo** column specifies if the speaker is part of a certain parliamentary group. The EP dataset has numerous unique columns: **MEP** and **commission** indicate if the speaker is a member of the European parliament or the European commission respectively. **Written** is a logical variable holding whether the speech was submitted in written form. In these cases of written speeches, it is possible that the text was submitted by more than one person which is indicated with the variable **multispeaker**. The Hungarian and the EP dataset contain a variable **translatedText** with English translations of the speech column. As not all speeches in the EP dataset were translated, the variable **translationInSpeech** indicates cases with translation.

3.2) Bills

3.2.1) Core variables

The second dataset holds the cleaned full-text vectors of bills (**bill_text**), as well as their titles (**title_bill**) and relevant metadata (Table 3). The legislative body's final decision regarding the bill is stored in the column **status**, making the comparison of, for example, rejected and passed bills possible. The **bill_ID** variable adheres to each parliament's naming conventions, enabling researchers to match bills with external data sources beyond the PLS corpora. Finally, similarly to the speeches' dataset, the **initiation_date** (in YYYY-MM-DD format) pinpoints the introduction date of each bill.

Bills corpora	
<i>Data on legislative bills tabled in the respective parliament</i>	
Variable	Description
title_bill	The title of the bill
bill_text	Full text of the bill
status	The final decision of the legislative body regarding the bill
procedure_ID	The unique PLS identifier linking speeches, bills, and laws
initiation_date	The day on which the bill was introduced

Table 3: Core variables of the bills corpora¹

3.2.2) Additional variables

There are several extra variables in the bills corpora across the legislatures. **Bill_ID** is a identification number of the bill, following the conventions of the respective parliaments. The **period** column clarifies the legislative period within which the bills were introduced. Tracking the legislative activity of different members of the parliament (and indirectly of parties, for instance), the **initiator** of the given bills is also included. Bill types are specified in the **bill_type** column (available for two countries), categorising them according to the respective parliaments' systems (such as by the initiators in the Spanish case). In the case of Croatia, bills and proposals stemming from EU law are indicated with the variable **bill_EU**.

Information regarding the legislative process (such as debates and decisions on the bill introduced) is also included in certain cases. **Voting_result & amendments** provide additional angles for researchers, possibly aiding with the recreation of comparing the unfolding of the legislative processes. The **committee** variable indicates whether the bill was referred to committees for consideration, offering insight into the black box of governance. The **bill_link** for the bill texts are also included in some cases.

¹ The variable status is not available in the Czech Republic corpus.

The unique characteristics of EP bills also appear in the columns: **CELEX**, **procedure_ID**, and **proposal_number** hold an identifier assigned to each respective document (instead of the variable bill_ID). **Procedure_type** specifies the type of legislative procedure under which the bill falls.

3.3) Laws

3.3.1) Core variables

The third corpora, comprising full-text vectors of laws (**law_text**) and their titles (**title_law**), is completed with metadata and presented in Table 4. The variables in these datasets resemble that of the speeches and bills corpora: besides the **procedure_ID**, **adoption_date** provides a picture of the legislative timeline. It is noteworthy that the length and content of bill and law titles may vary both across and within legislatures due to differences in legislative practices or drafting conventions.

Laws corpora	
<i>Data on laws finally adopted by the respective parliament</i>	
Variable	Description
title_law	The title of the law
law_text	The full text of the law
adoption_date	The day on which the law was published
procedure_ID	The unique PLS identifier linking speeches, bills, and laws

Table 4: Core variables in the laws corpora

3.3.2) Additional variables

The unique columns to these corpora include several useful additions, such as the legislative **period**, available for researchers concentrating on wider horizons. **Law_IDs** cater for easier identification when matching with external data, while the **law_links** are provided for traceability. The EP's **law_type** variable is analogous to the **bill_type** variable of the bill dataset.

4. Validation

A custom R script was developed to facilitate manual validation of critical aspects within the corpora. One key objective was to assess overall data completeness and identify potential duplicate and missing entries. Individual inspection of the data was necessary due to the possibility of encountering identical texts originating from distinct speeches.

Next, the number of observations for each scraped database (or subset) was verified by consulting the corresponding source websites. Additionally, a random sampling approach based on predefined criteria was implemented to further assess data representativeness. Ideally, data completeness was corroborated using an independent data source, such as alternative websites (e.g., party websites) or supplementary channels within the parliamentary websites themselves.

The metadata columns underwent a comprehensive cleaning process to ensure the clarity and consistency of both values and labels. Missing data points (NAs) were identified, and their potential for inclusion in the subsequent analysis was evaluated based on domain knowledge and data distribution patterns.

To ensure the quality of text data for subsequent use and analyses, a meticulous cleaning process was undertaken for each corpus. This iterative process involved the extraction of random text samples, followed by manual identification and rectification of any error patterns found within the samples. Examples of errors addressed included incorrect encoding (ensuring UTF-8 unless otherwise specified), missing headers or footers, and incomplete text. The iterative process continued until no further error patterns were detected within the random samples, suggesting a high level of data cleanliness.

5. Data access

ParlLawSpeech operates within an open science framework, with all data being openly published on public repositories. Access is granted based on the [CC-BY 4.0 Deed licence](#), which is the sole official PLS version.

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Appendix

ParlLawSpeech

Data Validation Workflow

This document outlines the steps taken to validate the data collected in the ParlLawSpeech Corpora. The validation process aims to ensure the completeness, consistency, and quality of the data across multiple corpora (laws, bills, speeches). Below, we provide an overview of the general validation workflow and the specific steps applied.

Overview of Validation Steps

1. General Completeness and Duplicate Checks
 - a. Verify the uniqueness of data points (e.g., IDs, text, links).
 - b. Test whether all targeted data points have been successfully scraped.
 - c. Compare scraped data against alternative sources to validate completeness.
2. Metadata Inspection
 - a. Check metadata columns (e.g., party names, speaker roles, dates, IDs) for errors or inconsistencies.
3. Cross-Dataset Linking
 - a. Ensure the datasets (bills, laws, speeches) can be linked through shared IDs and validate the consistency of these links.
4. Text Content Validation
 - a. Validate the quality and completeness of text data (speeches, bills, laws) through an iterative process.
5. Exploratory Validation
 - a. Identify anomalies in text length, punctuation, or structure using descriptive statistics and visualizations.
 - b. Leverage automated text analysis methods (e.g., sentiment analysis, topic modeling) to detect unusual patterns.

1. General Completeness and Duplicate Checks

- a) Verify the uniqueness of data points (e.g., IDs, text, links).

We start by looking for general inconsistencies such as duplicate cases, or missing data points. In the very first step, we verify whether columns that should be unique (ID, text, link, etc.) contain no duplicates. Note that whether a column is expected to or should be unique strongly depends on the context. For instance, duplicate speech texts can be *true* duplicates (e.g., short responses such as “Thank you.”). Other cases (e.g., duplicate bill texts) require manual validation.



b) Test whether all targeted data points have been successfully scraped.

Duplicate checks are followed by testing the completeness of the scraped corpus. Validating the performance of the scraper in terms of completeness is a delicate task, as we must ensure that errors in our scraper do not translate into our validation method. Therefore, ideally the completeness of the scraped data is validated through both the original and alternative data sources or through alternative means. Alternatives could for instance be different web sources (e.g., party websites) or different channels on the original source / parliamentary websites (e.g.; filtering by MP, committee,...; search via the parliamentary “media library” instead of “data documentation”). The final selection of (combinations of) validation data per country largely depends on the availability of alternative sources and might therefore strongly differ by countries. However, the general validation process, once the validation source has been identified, is largely the same. After identifying the target number of observations through original and alternative data sources, we validate whether the number of observations in the scraped corpus matches the target number of observations overall and by subsettable groups (e.g., speaker name, committee, party, month, document type, ...).

e.g., in Germany, the following alternative sources were identified:

speeches

- [Deutscher Bundestag - Mediathek](#) (complete videos of all speeches made)
=> it is possible to compare dates, agendas, text content between media library and scraped texts
- Sometimes the party websites do have the speeches of their members; this is not always the case however, so a check with party websites alone is very spotty
=> die Grünen: [Bundestagsreden \(gruene-bundestag.de\)](#)
=> die Linke: [Reden - Fraktion DIE LINKE. im Bundestag \(linksfraktion.de\)](#)

bills/laws

- [DIP](#) (“Dokumentations- und Informationssystem für Parlamentsmaterialien”)
- [Bundestagsgesetzblatt](#) (all published laws are collected here)

c) Check if all texts have been extracted.

In this step, we confirm whether all expected texts have been extracted from the source. This involves checking that no entries are missing (i.e., != NA). However, this step only confirms the existence of text data; the completeness of the text content itself is addressed in Step 4.



2. Metadata Inspection

- a) Check metadata columns (e.g., party names, speaker roles, dates, IDs) for errors or inconsistencies.

This step focuses on ensuring the accuracy and consistency of metadata across the dataset. Validation includes:

- **Standardization of Text Fields:** Verify that categorical metadata (e.g., party names, speaker roles) is correctly labeled and consistent. For example, party names should match their official titles without variations or spelling errors.
- **Validation of IDs:** Check that IDs (as retrieved from the data source) follow the expected format and structure specific to the source.

Semi-automated tools such as regex can identify irregularities in structured fields like IDs, enabling efficient detection and correction of errors. Adjustments are applied iteratively to ensure consistency throughout the dataset.

3. Cross-Dataset Linking

- a) Ensure the datasets (bills, laws, speeches) can be linked through shared IDs and validate the consistency of these links.

This step ensures that the different datasets—bills, laws, and speeches—are correctly linked via the ID column, enabling further analyses. Validating these links is crucial to maintain consistency across corpora and identify any potential discrepancies.

For the bill and law data, validation involves performing a full join between the two datasets. The number of observations in the joined dataset should match the total number of bills. It is important to note that the texts of bills and laws should not overlap, except in cases where both texts are extracted from the same document, as this is sometimes the case in certain countries. When texts are extracted from the same document, their titles are expected to match.

For the relationship between bills and speeches, it is necessary to account for the fact that multiple bills can be associated with a single speech, and multiple speeches may reference the same bill. In the data, this relationship is managed by separating the IDs of multiple bills referenced in a single speech with commas (e.g., 19/50673/19, 19/4562/19). These entries are then unnested—converted from wide to long format—so that each bill ID has its own entry in the dataset. After unnesting, duplicate IDs are removed, ensuring that each bill ID appears only once in the speech-to-bill linking process.



Following this preparation, the total number of bills is compared to the number of rows in the joined dataset. If the joined dataset contains more rows than the number of bills, this indicates the presence of IDs in the speech dataset that do not match any IDs in the bill dataset. Such discrepancies suggest errors in the IDs that require further investigation and correction. Some country datasets contain bills that are not debated in parliament and are therefore not associated with speeches. Some of these unassociated bills later become laws, which can be verified by linking the bill IDs to the law data.

4. Text Content

a) Descriptive Plots for Outlier Detection

This step ensures the quality and completeness of the text data (speeches, bills, laws) through an iterative process. The validation involves extracting random samples, identifying and correcting errors, and testing for tokenization accuracy. These steps are repeated until no more error patterns are identified in a random sample of at least 10 texts.

Step 1: Sampling

A random sample of text data is extracted from the dataset, with each sampled row saved in its current processed and raw forms as local text files. The sampling process keeps track of previously selected texts to ensure that no row is sampled more than once.

Step 2: Identify & correct errors

After sampling, errors in the text data are identified and corrected. This process involves comparing processed texts, raw files, and original documents to ensure that the cleaned texts are accurate and complete representations of the originals. Key focus areas include:

- a) Encoding: Ensuring characters are correctly encoded (e.g., UTF-8).
- b) Cleaning: Verifying that headers, footers, and other extraneous content have been removed. Note that this can be particularly challenging in certain cases (e.g., Hungary).
- c) Completeness: Checking for completeness by reviewing patterns at the start and end of documents.
- d) Targeted Content: Confirming that the text includes only the intended content. Errors, such as multiple target elements (e.g., two speeches in one), can be identified by skimming paragraphs in the sampled text.
- e) Hyphenation: Ensuring hyphenated words have been correctly identified and de-hyphenated.

Errors that stem from inaccuracies in the original documents cannot be corrected. However, for errors introduced during preprocessing, adjustments are made to the corresponding code, typically in the script that processes raw texts into clean texts.

Step 3: Save Progress and Restart Cycle

After each iteration, the current state of all sampled texts is saved, with filenames containing the row number and iteration number. This allows for tracking progress and identifying whether new errors are introduced during subsequent corrections.



Step 4: Test for Sentence Tokenization After Cleaning

To validate sentence tokenization, a sample of bills, laws, and speeches is selected. The number of sentences in the cleaned text is counted and compared to the count derived from `str_count`. Any discrepancies can help identify lingering issues with tokenization.

5. Exploratory Validation

As a final step, potential errors in the dataset are identified through outlier analysis and other diagnostic techniques. This process involves creating descriptive visualizations and leveraging automated text analysis methods to detect anomalies that may indicate underlying issues.

- a) Identify anomalies in text length, punctuation, or structure using descriptive statistics and visualizations.

Basic scatterplots and visualizations are used to identify inconsistencies in the dataset. These plots highlight relationships between key text features, and deviations from expected patterns can signal potential errors. Examples include:

- Scatterplots of text length (word count) by punctuation count: This should show a linear correlation. Significant outliers may point to incorrectly processed or incomplete text.
 - Scatterplots of cleaned text length by raw text length: A linear relationship is expected; unusually short cleaned texts may indicate issues during the cleaning process.
 - Scatterplots of text length by number of sentences: If sentence separation is possible, this plot can help evaluate the accuracy of tokenization and identify unusual cases.
 - Scatterplots of bill text length by law text length: A linear correlation is generally expected, with outliers potentially reflecting mismatched or incomplete data.
- b) Leverage automated text analysis methods (e.g., sentiment analysis, topic modeling) to detect unusual patterns.

Automated techniques provide additional insights into potential data quality issues by examining text content in aggregate. Examples include:

- Sentiment Analysis: Identifies unexpected sentiment patterns that may indicate misclassification or other anomalies.
- Wordfish Analysis: Highlights unusual textual patterns that could stem from errors in text processing.
- Unsupervised Topic Modeling: Detects unusual or irrelevant topics that may indicate text contamination or mislabeling.
- TF-IDF Matrix Comparisons: Highlights tokens with unusually low scores compared to other corpora, which may signal discrepancies or missing content.

